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Analysis of Antibiotic Resistance and Virulence Patterns in *Klebsiella Pneumoniae* Isolated from Human Urinary Tract Infections in India

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Significance and impact of the study

We surveyed antimicrobial resistance trends in *Klebsiella pneumoniae* recovered from urinary tract infection (UTI) patients in Krishna district, Andhra Pradesh, India. This study revealed alarming levels of resistance against multiple antibiotics classes. To circumvent this, hospitals need to implement strict antimicrobial susceptibility testing protocols before initiating the treatment to patients. Comparison of antimicrobial resistance patterns in same region at different time points will aid in assessing the effectiveness of interventions, establishing antibiotic resistance (ABR) trends in pathogens and identifying emerging pathogens.

Abstract

In this study, 504 urine samples collected from patients suffering from UTIs were screened for presence of *K. pneumoniae*. The overall occurrence of *Klebsiella spp.* and *K. pneumoniae* was found to be 23.2% (117/504) and 16.8% (85/504) respectively. Antibiotic susceptibility testing of 85 *K. pneumoniae* isolates was carried out by disc diffusion which revealed alarming levels of antibiotic resistance. Antimicrobial resistance was prominently observed against cefpodoxime (76.47%) followed by ampicillin (70.59%), ceftriaxone (52.94%), ceftiofloxacin (50.59%), amoxycyclav (48.24%), ofloxacin (45.88%), cefotaxime (44.71%), cefepime (43.53%) and doxycycline hydrochloride (40%). A small percentage of strains also exhibited resistance to other antimicrobials in the range of 7 – 35 %. Around 77.6% of the isolates were found to be resistant to three or more antibiotic classes and 66.7% of the isolates had multiple antibiotic resistance (MAR) index values >0.2. Screening of virulence genes in 85 *K. pneumoniae* isolates revealed that *uge* gene was the most predominant (11/85, 12.9%), followed by *rmpA* (9/85, 10.5%), *kfu* (4/85, 4.7%) and aerobactin genes (2/85, 2.35%). Further, the overall percentage of biofilm producers were found to be 17.65% (15/85). This study warrants hospitals and health care centres to reduce misuse of antibiotics and manage UTI with appropriate treatment after performing antibiotic susceptibility testing.

Keywords

Klebsiella pneumoniae, Urinary tract infections, Multidrug resistance, Disk diffusion assay, Biofilm, Virulence profile

Introduction

Emergence and dissemination of ABR mechanisms is a major public health concern globally which put the health care sector at a greater disadvantage (Shaikh *et al.* 2016; Mbelle *et al.* 2020). ABR is associated with treatment failures, increased mortality and morbidity. World Health Organization (WHO) has classified the cephalosporin- and carbapenem-resistant *Enterobacteriaceae* members as critical priority pathogens (Tacconelli *et al.* 2017). Among these, *Klebsiella pneumoniae* is notorious for rapid development of ABR. *K. pneumoniae* ranks second among the causes of opportunistic nosocomial Gram-negative bacteraemia among immunocompromised patients (Zheng *et al.* 2018). *K. pneumoniae* colonizes skin and mucosa, and causes UTIs, septicaemia and pneumoniae (Skowron *et al.* 2019). Resistance to different antimicrobial classes such as third-generation cephalosporins, aminoglycosides and quinolones in *K. pneumoniae* have been mentioned in various studies (Alcantar-Curiel *et al.* 2018). The mechanism implicated for the dissemination of resistance is horizontal transmission mediated by mobile genetic elements (MGE) such as plasmids, transposons and integrons.

Virulence markers help us to elucidate the mechanism of pathogenesis of *K. pneumoniae* within the body of the host. Some of the important virulence genes are regulator of mucoid phenotype A (*rmpA*) and uridine diphosphate galactose 4 epimerase (*uge*) responsible for capsule and smooth polysaccharide synthesis, aerobactin (a hydroxamate-type siderophore iron-uptake system) and *kfu* (iron uptake system) (Osman *et al.* 2014; Zhang *et al.* 2018). They work individually, synergistically or in tandem to orchestrate the pathogenesis within the host (Osman *et al.* 2014). Biofilm formation is another crucial virulence mechanism of this organism responsible for its nosocomial importance. Biofilm is a group of microorganisms that are attached to living and non-living surface and plays a protective role against antibiotics and macrophages (Yazgan *et al.* 2018). Ever since the first report of biofilm formation phenomenon in *K. pneumoniae* emerged in 1988, many research works have suggested a strong association between biofilm formation, virulence and multidrug resistance (MDR) mechanisms (Nirwati *et al.* 2019). *K. pneumoniae* utilize adhesins and exopolysaccharides to grow in sessile mode and employ multiple mechanisms such as efflux pumps, quorum sensing and LPS production to produce a complex biofilm. These mechanisms are interconnected and are relevant in

ABR and biofilm formation and thus render *K. pneumoniae* strains more virulent and ABR while growing as biofilm (Vuotto *et al.* 2017; Zhang *et al.* 2018; Pelling *et al.* 2019). Biofilm mechanisms have severe clinical implications in antimicrobial chemotherapy, disease prognosis and disease control. Biofilm-forming bacteria have reduced susceptibility to antimicrobials, due to exclusion of the antimicrobial and slow bacterial growth rates that impede antagonistic activity (Klima *et al.* 2017). Biofilm phenomenon is responsible for hospital-associated UTI which accounts for approximately 40 % of the total nosocomial infections in the United States (Saint and Chenoweth 2003).

The present study focussed on the prevalence of *K. pneumoniae* in human UTIs in Krishna District of Andhra Pradesh, India. There are no previous studies undertaken on this subject in this region. Harboursing antibiotic resistance, virulence genes and the ability to form biofilm dictates the virulence status of the strain and thereby the severity of the infection. The aim of this study is to evaluate the percentage distribution of antibiotic resistance, patterns of MAR in *K. pneumoniae* isolates and their ability to form biofilms *in vitro*.

Results and discussion

Isolation and characterization

The overall occurrence of *Klebsiella* spp. in urine samples was found to be 23.21% (117 isolates) and that of *K. pneumoniae* was 16.86% (85 isolates) based on the PCR result for *gyrA* and *rpoB* genes for genus and species identification respectively. The isolation rate of *K. pneumoniae* was closely comparable with the findings of other researchers. For example, in the study by Manjula *et al.* 2014 they had reported 9.8 % isolation rate of *K. pneumoniae* from pregnant female UTI patients. This study was conducted in Karnataka state which is the border state of Andhra Pradesh. In contrast, in another study by Abadulla and Zghair 2016, they reported a low (1.9 %) *K. pneumoniae* isolation rate from human urine samples. This study was undertaken in Baghdad city, Iraq. Zhang *et al.* 2018 reported a net 7.8 % of *K. pneumoniae* isolation rate from stool samples of diarrhoeal patients from

Tai'an city, China. Difference in isolation rate could be due to geographical, seasonal variations and sample types.

Antibiotic susceptibility testing

High prevalence of MDR strains of *K. pneumoniae* is a serious public health concern (Li *et al.* 2014; Dey *et al.* 2020). In India, limited studies are available on prevalence of multidrug-resistant *K. pneumoniae* from hospitals and community. In the present study, all the 85 *K. pneumoniae* isolates were subjected to antibiotic susceptibility testing against seventeen antibiotics (Figure 1A). Susceptibility testing revealed highest resistance against cefpodoxime followed by ampicillin, ceftriaxone, ceftiofur, amoxycylav, ofloxacin, cefotaxime, cefepime, and doxycycline hydrochloride. Only one isolate was found to be resistant to meropenem. The resistance levels towards cephalosporins ranged from 35.29 % to 76.47 %. The resistance level against gentamicin was observed at 21 % whereas the resistance level toward nitrofurantoin was 30.6 %. *K. pneumoniae* isolates showed highest susceptibility to meropenem at 98.8 %, amikacin at 93 %, and piperacillin/tazobactam at 90 % some of which could be in use in clinic for treatment. Several groups have reported the percentage ESBL-producing *K. pneumoniae* at 4 % to 83 % in India (Shukla *et al.* 2004; Sarojamma and Ramakrishna 2011). In this study, the percentage of ESBL-producing *K. pneumoniae* were found in the range of 35 % to 76 %. *K. pneumoniae* frequently produce ESBLs which confer resistance to β -lactam antibiotics including 4th generation cephalosporins and monobactams (Pana and Zaoutis 2018). ESBL-producing *K. pneumoniae* causing UTI in children and adults is a serious concern worldwide which progresses to sepsis and prolonged hospitalizations. Isolation of ESBL-producing *K. pneumoniae* severely limits therapeutic options and the treatment are often delayed due to antibiotic susceptibility testing (Vachvanichsanong *et al.* 2021). ESBL-producing strains of *K. pneumoniae* are widespread across the world causing numerous outbreaks (Tumbarello *et al.* 2006). Under such circumstances, carbapenems represent the first line of defence against ESBL-producing *K. pneumoniae*. However, carbapenem-resistant *K. pneumoniae* (CRKP) have been on the constant rise receiving wide spread attention due to restricted limitations antimicrobial treatment (Xu *et al.* 2017). Although the resistance against meropenem is very limited (1.18 %) in

this study, still it deserves attention for appropriate measures. Results of antibiotic susceptibility testing against the 17 antibiotics are presented in table 1.

Table 1 Percentage of antibiotic resistance levels observed in *K. pneumoniae* isolates

ANTIBIOTICS	Number of resistant isolates*	% of RESISTANT STRAINS
Amikacin (AK-30 µg)	06	7.06
Amoxyclav (AMC-20/10 µg)	41	48.24
Ampicillin (AMP-10 µg)	60	70.59
Aztreonam (AT-30 µg)	27	31.76
Cefepime (CPM-30 µg)	37	43.53
Cefotaxime (CTX-30 µg)	38	44.71
Cefotetan (CTN-30 µg)	30	35.29
Cefoxitin (CX-30 µg)	43	50.59
Cefpodoxime (CPD-10 µg)	65	76.47
Ceftazidime (CAZ-30 µg)	30	35.29
Ceftriaxone (CTR-30 µg)	45	52.94
Doxycycline Hydrochloride (DO-30 µg)	34	40.00
Gentamicin (GEN-10 µg)	18	21.18
Meropenem (MRP-10 µg)	1	1.18

Nitrofurantoin (NIT-300 µg)	26	30.59
Ofloxacin (OF-5 µg)	39	45.88
Piperacillin/Tazobactam (PIT- 100/10 µg)	8	9.41

* N = 85 (N – Total number of *K. pneumoniae* tested)

Majority of *K. pneumoniae* isolates recovered in this study are MDR organisms limiting the available treatment options. Some of these isolates simultaneously showed resistance to multiple antibiotic classes implying that uncontrolled antibiotic usage causes selection of ABR, co-selection of resistance genes through horizontal gene transfer.

MAR indexing

Resistance was observed among isolates towards multiple antibiotic classes. Around 77.65 % of the isolates were found to be resistant to three or more classes of antibiotics. The MAR index values were appropriately calculated and illustrated in the figure 1B. Around 78.82 % of the isolates have MAR values above the cut-off value of 0.2 which is suggestive of exposure to high antibiotic environment. Each isolate was resistant to 3–15 antibiotics. Therefore, strict monitoring of consumption of 3rd and 4th generation cephalosporins and carbapenems combined with frequent tracing of evolutionary trends in susceptibility and resistance patterns of *K. pneumoniae* in the community-acquired infections is highly mandatory. The present study suggests that meropenem, amikacin and piperacillin/tazobactam should be considered to be a viable treatment option for treatment of difficult to treat MDR UTI infections.

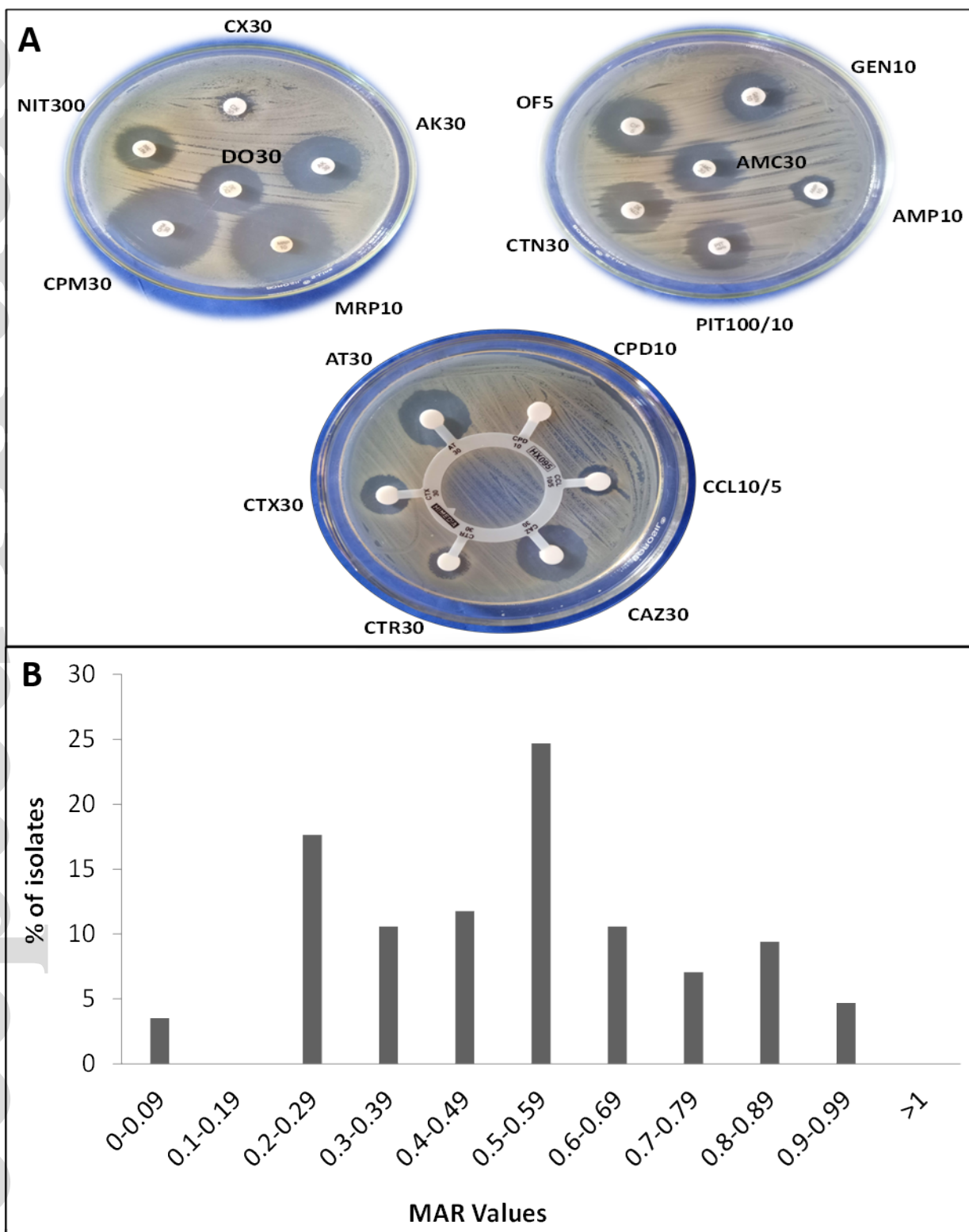


Figure 1 (A) Kirby Bauer antibiotic disk diffusion test for screening the resistance and sensitivity of *K. pneumoniae* isolated from urine of UTI patients. CX30 – Cefoxitin 30 µg; AK30 – Amikacin 30 mcg; MRP10 – Meropenem 10 mcg; CPM30 – Cefepime 30 mcg; NIT300 – Nitrofurantoin 300 mcg; DO30 – Doxycycline hydrochloride 30 mcg; GEN10 – Gentamicin 10 mcg; AMP10 – Ampicillin 10 mcg; PIT100/10 – Piperacillin/Tazobactam 100/ 10 mcg; CTN30 – Cefotetan 30 mcg; OF5 – Ofloxacin 5 mcg; AMC30 – Amoxicillin/Clavulanic Acid 30 mcg; CPD10 – Cefpodoxime 10 mcg; CCL10/5 Cefpodoxime/Clavulanic acid 10/5 mcg; CAZ30 – Ceftazidime 30 mcg; CTR30 – Ceftriaxone 30 mcg; CTX30 – Cefotaxime 30 mcg; AT30 – Aztreonam 30 mcg. (B) MAR index levels observed in *K. pneumoniae* isolates. Approximately 79 % of isolates have MAR values above the cut-off value of 0.2.

Virulence gene profiling

Molecular detection of virulence genes in *K. pneumoniae* isolates revealed the presence of four virulence markers with *uge* gene being the most predominant (11/85, 12.9%), followed by *rpmA* (9/85, 10.5%), *kfu* (4/85, 4.7%) and aerobactin genes (2/85, 2.35%) (Figure 2). Remya *et al.* (2019) studied the prevalence of virulence genes in 370 ESBL-positive *K. pneumoniae* isolates recovered from clinical samples and reported that *uge* gene (48.65%) predominated over *kfu* (27.84%), aerobactin (5.4%) and *rpmA* (5.13%). They could not detect any virulence genes in 2.7% of their isolates. The levels of virulence genes were comparable with previous works by different authors where a wide range of values was observed and reported (Ali *et al.* 2019; Shakib *et al.* 2018; Jian-Li *et al.* 2017). However, the difference among the values could be due to different geographical location and sample size.

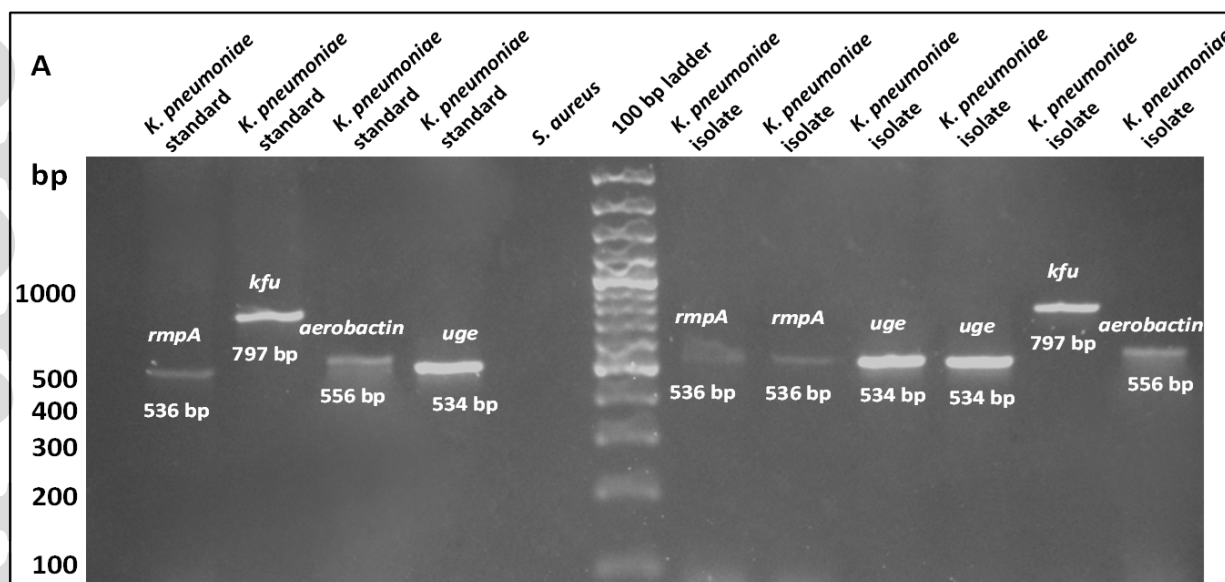


Figure 2 (A) Agarose gel showing the PCR amplified virulence gene products of *K. pneumoniae*. *rmpA* (regulator of mucoid phenotype A) gene plays major role in polysaccharide synthesis and virulence. *kfu* gene product mediates uptake of iron, aerobactin is a hydroxamate siderophore, *uge* codes for UDP galacturonate 4-epimerase plays role in capsular polysaccharide synthesis. These genes are typically present in hypervirulent *K. pneumoniae* strains.

Biofilm formation

Biofilm formation was observed in 17.65 % of the isolates. The positive isolates were subjected to MTP assay to ascertain the strength which revealed that strong, medium and low biofilm-forming ability were observed in 3 (20%), 5 (33.4%) and 7 (46.7%) isolates respectively. Seifi *et al.* (2016) conducted a study on biofilm-forming ability among *K. pneumoniae* isolates from clinical specimens and reported that among the 56 urine isolates, 21.42%, 71.43% and 7.14% exhibited strong, moderate and weak biofilm-forming ability, respectively. The difference in the values can be attributed to various physical, chemical and geographical sampling factors. Cepas *et al.* 2019 have reported that acquisition of specific ABR can lower or enhance the biofilm-forming ability variedly in different Gram-negative bacterial pathogens. They postulated that MDR isolates need not necessarily be a copious biofilm former in comparison to non-MDR isolate. On the contrary, in this

study a few of the MDR *K. pneumoniae* isolates harboured virulence genes and demonstrated the strong capacity for biofilm formation indicating the pathogenic nature of these strains. Many studies have linked biofilm-forming ability with increased ABR, ESBL production and higher virulence. In one study, a significant association was discovered between biofilm formation and ESBL production in 150 *K. pneumoniae* isolates from sputum and urine (Yang and Zhang 2008). In another study by Subramanian *et al.* 2012 they have discovered biofilm positive isolates displaying more resistance to nalidixic acid, ampicillin, cefotaxime and co-trimoxazole in comparison to biofilm nonproducers. Sanchez *et al.* 2013 have observed *K. pneumoniae* with MDR phenotype capable of forming richer biofilm than the susceptible ones. We presume that ABR mechanisms, virulence genes and biofilm formation are interconnected and may confer the strains with higher virulence and antibiotic resistance while growing as biofilm. These phenomena have clinical implications in disease progression, antimicrobial treatment and disease outcome.

There is an overall trend of variable spike in prevalence of MDR *K. pneumoniae* strains across different parts of India. The present study reports the prevalence of large number of MDR *K. pneumoniae* from UTIs in Krishna district of Andhra Pradesh. In this study, in spite of high prevalence of MDR strains and MAR indexes, the biofilm formation was observed in relatively fewer strains. A better understanding of the action of virulence genes and the biofilm formation is necessary to break the chain of infection. In India, indiscriminate prescription of antibiotics and patients lack of compliance is common which could be the reason for MDR phenomenon (Prakasam *et al.* 2012). *K. pneumoniae* isolates showed high resistance to commonly used antibiotics, and they were susceptible only to small number of antibiotics such meropenem, amikacin and piperacillin/tazobactam and therefore these antibiotics could be choice of treatment for complicated UTIs. Even though, this study identified only one isolate resistant to meropenem, it calls for an urgent evaluation of its use in the treatment of UTIs. In addition, this study has some limitations such as *K. pneumoniae* small isolate size, lack of screening of virulence genes coding for capsular serotype, adhesins, liposaccharides, iron acquisition system etc, and thorough correlation between ABR, virulence genes and biofilm-forming ability is needed in future studies. Further,

routine screening of ABR microorganisms is needed to be undertaken by each hospital/ laboratory before prescribing proper therapeutic regimen. Hospitals and diagnostic centres should lay emphasis on carrying out compulsory disk diffusion test mixed with rationale prescription of available antibiotics minimizing the misuse followed by undertaking regular antibiotic susceptibility surveillance studies to ascertain the efficacy of interventions.

Materials and methods

Materials and standard strains

Dehydrated microbial culture media, supplements and antibiotic discs were procured from HiMedia laboratories, India. Molecular biology grade agarose, trizma, boric acid and Ethylenediaminetetraacetic acid (EDTA) were procured from Sigma Aldrich, India. PCR master mix was procured from Promega, USA. The American Type Culture Collection (ATCC) reference strain, *K. pneumoniae* ATCC 700603 was taken as standard. The oligos were procured from Bioserve Biotechnologies Pvt. Ltd., Hyderabad, India.

Sample Collection

The study was conducted during the period from April, 2017 to February, 2020. A total of 504 human urine samples collected from UTI patients in different diagnostic labs in Krishna district of Andhra Pradesh were screened in this study. Patients with UTI belonging to both genders were included in the study. All the patients involved in the study were under the age group of 15 – 64 years old. Patients included in the study did not have any other coinfections apart from UTI. Institutional Biosafety committee (IBSC) of Vignan's Foundation for Science, Technology and Research approved the study design and protocols employed. Informed consent was taken from patients to use the urine sample for *Klebsiella* isolation studies. Once the samples were brought to laboratory, they were immediately processed by incubating 1 mL in 9 mL of trypticase soy broth at 37°C for 24 h. The enriched inoculum was streaked on MacConkey agar (MCA) plates and incubated at 37°C. Identification and confirmation were done phenotypically by colony morphology, staining

characteristics and biochemical tests as well as molecular characterization by PCR (Cruickshank 1980; Chander *et al.*, 2011).

Genotypic confirmation by PCR

Genomic DNA extraction was performed by boiling and chilling method (Bobbadi *et al.* 2019). Molecular confirmation was done by a duplex PCR targeting genus and species-specific regions (Table 2). PCR assay was optimized in 25 µL reaction mixture containing 1X master mix (GoTaq Promega Green Master Mix), 5 pmoles each of forward and reverse primers (10 pmol/µL), 10 – 20 µg of genomic DNA and the final volume was made up by adding nuclease free water. The cycling conditions were standardized at initial denaturation of 95 °C for 5 min, 30 cycles of denaturation at 95°C for 1 min, annealing at 55°C for 1 min and extension at 72°C for 1 min followed by final extension at 72°C for 10 min. PCR products were subjected to electrophoresis on 1.5 % agarose gel with 2.5 µg/mL of ethidium bromide as fluorescent dye (Sambrook and Russell 2001) and visualized using gel documentation unit (BIORAD, USA).

Table 2: List of primers used for the detection of *K. pneumoniae* isolates and assessment of virulence gene repertoire of *K. pneumoniae* isolates

TARGET GENE	PRIMER SEQUENCE (5'-3')	AMPLICON SIZE (bp)	REFERENCE
Genus and species-specific primer for identification of <i>K. pneumoniae</i>			
<i>gyrA</i>	CGC GTA CTA TAC GCC ATG AAC GTA	441	Chander et al. 2011
	ACC GTT GAT CAC TTC GGT CAG G		
<i>rpoB</i>	CAA CGG TGT GGT TAC TGA CG	108	
	TCT ACG AAG TGG CCG TTT TC		

Primer pairs for profiling *K. pneumoniae* virulence genes

<i>rmpA</i>	ACT GGG CTA CCT CTG CTT CA	536	Osman et al. 2014
	CTT GCA TGA GCC ATC TTT CA		
<i>Uge</i>	TCT TCA CGC CTT CCT TCA CT	534	
	GAT CAT CCG GTC TCC CTG TA		
<i>Kfu</i>	GAA GTG ACG CTG TTT CTG GC	797	
	TTT CGT GTG GCC AGT GAC TC		
Aerobact in	GCA TAG GCG GAT ACG AAC AT	556	
	CAC AGG GCA ATT GCT TAC CT		

Antibiogram profiles and Multiple Antibiotic Resistance (MAR) Indexing

All the confirmed isolates were subjected to antimicrobial susceptibility testing by exposing them to 17 antibiotics (cefpodoxime, ampicillin, ceftriaxone, cefoxitin, amoxiclav, ofloxacin, cefotaxime, cefepime, doxycycline, meropenem, amikacin, gentamycin, nitrofurantoin, aztreonam, cefotetan, ceftazidime and piperacillin/tazobactam) belonging to different classes. Kirby-Bauer disc diffusion method was employed with lawn culture on Mueller-Hinton agar (Bauer *et al.* 1966). Direct colony suspension of each isolate was made in PBS (pH 7.4) and the turbidity was adjusted to 0.5 McFarland units (equivalent to an approximate cell density of 1.5×10^8 CFU/mL). The diameter of inhibition zones was measured and susceptibility patterns of *Klebsiella* species were interpreted according to Clinical and Laboratory Standards Institute (CLSI) guidelines (CLSI 2014). Isolates that showed resistance to three or more antibiotic classes were considered as MDR strain (Magiorakos *et al.*

2012). The MAR index values for the isolates were calculated as the proportion of antibiotics resistant to the total number of antibiotics used (Srinivas *et al.* 2019). The formula is given as:

MAR index of an isolate = number of antibiotics resistant to/ number of antibiotics used

Virulence gene profiling

Molecular characterization of virulence genes was performed with the help of one monoplex for *rmpA* gene and one multiplex PCR assay targeting three genes (*uge*, *kfu*, aerobactin). PCR assays were optimized for 25 µl reactions and the PCR cycling conditions and primers sequences were used as per the work of Osman *et al.* 2014 (Table 2).

Biofilm formation

Biofilm-forming ability was identified qualitatively and quantitatively by Congo Red agar (CRA) and microtiter plate (MTP) assays respectively. CRA detection of biofilm was performed as per the protocol laid down by Freeman *et al.* (1989). CRA was prepared by supplementing BHI agar with 5 % sucrose and 0.08 % Congo red stain. Isolated strains were streaked on CRA plates and incubated at 37 °C for 24 h. *K. pneumoniae* isolates displaying crystalline black coloured colonies were considered biofilm producers and colonies showing pink to white colonies were considered as non-biofilm formers.

Biofilm formation by *K. pneumoniae* isolates was quantitatively estimated using the method of Maldonado *et al.* 2007. For microtiter plate assay, overnight grown *K. pneumoniae* strains were vortexed thoroughly and the OD was adjusted with sterile LB broth (supplemented with 1 % glucose) to 0.5 (~1x10⁸ cells/mL). Two hundred microliters of diluted strains were aliquoted into sterile, polystyrene 96-well Nunc microtiter plates. The plates were incubated at 37 °C for 6 h. Twenty-five µL of 1 % crystal violet was added to wells and agitated for proper mixing. The wells were washed with sterile 1x PBS three times to remove nonplanktonic cells and unbound stain. Plates were allowed to dry at room temperature and 200 µl of ethyl alcohol was added and mixed for dissolving the bound crystal violet. The resulting liquid alcohol was mixed with 1.2 mL of ethyl alcohol and the

absorbance of mixture was measured at 540 nm. *K. pneumoniae* ATCC 700603 served as positive control and the sterile LB broth supplemented with 1 % glucose without the inoculum treated under same conditions served as negative control. Each strain was tested three times and the average OD after blanking was considered for classification of strains as high producers (OD > 0.5), moderate producers (OD 0.5 – 0.1) and low producers (OD < 0.1) (Maldonado *et al.* 2007).

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Conflict of interests

Authors declare no conflict of interests

Ethical approval

No experimental animals were used in this study

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